
Robotic Mudskipper for Assessing the Efficiency of Water-Hopping

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Abstract. Mudskippers exhibit a unique water-hopping locomotion, landing on the water surface to create a cavity that may transfer additional energy to subsequent jumps. Studies were conducted with a 65 mm long robot to mimic this behaviour and determine whether energy recapture allows hopping to be a more efficient means of transport. In each recorded trial the robot starts fully submerged, and motors activate in a three-pulse repeating pattern to create a jump. Parameter optimisation was performed across robot fin size, fin placement and motor activation timing to best replicate hopping. An INA219 current sensor recorded current and voltage, from which power was calculated, while position was tracked from video using DLTdv8. Power and velocity data were used to identify regimes where hopping is more efficient due to energy recapture. The hopping behaviour operates at $Re \approx 45,400$ and $We \approx 441$ based on water-surface impact velocity, with $Fr \approx 0.88$ from impact velocity to 0.90 from cycle-averaged velocity, indicating a regime in which inertial and gravitational forces jointly influence water cavity dynamics. These results show how utilising water cavity collapse during hopping can reduce total travel energy compared with swimming, offering insight for scalable, energy-efficient amphibious transport and exploration.

Keywords. bio-inspired robotics; amphibious locomotion; water entry; cavity dynamics; energy recapture; mudskipper; organismic physics

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Optimised configuration

Five parameters were swept to obtain repeatable hopping. The configuration below is the one carried into the efficiency trials.

Characteristic length is 40 mm. All trials were conducted in an enclosed water tank and recorded with a GoPro at 240 fps, with a fishing-line guidewire constraining hops to a straight path and data processed in MATLAB.

Parameter	Selected value	Reason for varying
Voltage	19 V	Counteract gravity and downward momentum
Fin location	0.5 (mid-body)	Optimise vertical lift, reduce pitching
Fin size	$15 \times 30 \times 0.6$ mm	Balance slamming force against drag
Duty cycle	70 / 115 ms on-off	Thrust generation, hop rhythm, repeatability
Caudal fin	30 mm	Decrease rolling, increase hop count

Principal findings

- Across five trials each of hopping and underwater swimming, **hopping reduced total travel energy** relative to swimming over the same distance. Most of the saving is attributable to reduced drag in air; energy recapture from water cavity collapse may contribute but was not isolated.
- The current model **rolls**, which skews the forward trajectory and caps the achievable hop count. Reducing guidewire tension and replacing the original fin pair with a dihedral fin both improved stability.
- Drag enhancers were used to simulate higher speeds but degraded performance and reduced efficiency, and were excluded from the reported trials.

Next steps

Improve stability to reach three or more consecutive hops; adjust fin shape and centre of mass to assess their effect on hopping; and quantify the energy transferred from the cavity to the subsequent hop directly rather than by subtraction.

References

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Document note. Abstract as submitted to and accepted by the APS Global Physics Summit 2026 for poster presentation. The optimised configuration and principal findings from the accompanying poster are appended.